

Education

- 2021–Present **PhD in Quantum Information**, Freie Universität Berlin, Germany. *Advisor*: Jens Eisert.
Recipient of the [2025 Google PhD Fellowship in Quantum Computing](#), awarded to “outstanding graduate students doing exceptional and innovative research”.
- 2019–2021 **MSc in Physics** (Double Degree), University of Trento – SISSA, Italy. Graduated with 110/110 *cum laude* (highest possible grade); maximum average exam score.
- 2016–2019 **BSc in Physics**, University of Pisa, Italy. Graduated with 110/110 *cum laude* (highest possible grade).

Research Positions

- Summer 2026 **Graduate Student Researcher**, Google Quantum AI, Los Angeles, CA, USA. *Mentor*: Jarrod McClean.
- Summer 2024 **Graduate Student Researcher**, Los Alamos National Laboratory, NM, USA. *Mentor*: Marco Cerezo.

Publications ([Google Scholar](#), 1,600 citations, h-index 16; selected papers marked with ★)

1. ★ **A. A. Mele**, F. A. Mele, J. R. McClean, T. E. O'Brien. *Towards verifiable quantum advantage with random circuits: Observables that survive concentration*. [arXiv:2609.37890](#) (2026).
2. ★ **A. A. Mele**, L. Bittel. *Optimal learning of quantum channels in diamond distance*. *Nature Physics*, accepted for publication (2026). [arXiv:2512.10214](#).
3. ★ **A. A. Mele**, A. Angrisani, S. Ghosh, S. Khatri, J. Eisert, D. S. França, Y. Quek. *Noise-induced shallow circuits and the absence of barren plateaus*. *Nature Physics* 22, 751–756 (2026).
4. ★ F. A. Mele, **A. A. Mele**, L. Bittel, J. Eisert, V. Giovannetti, L. Lami, L. Leone, S. F. E. Oliviero. *Learning quantum states of continuous-variable systems*. *Nature Physics* 21, 2002–2008 (2025).
5. ★ S. Chen, **A. A. Mele**, F. A. Mele, J. Preskill. *When are bosonic Gaussian states classical to learn?* [arXiv:2609.26705](#) (2026).
6. ★ T. Parella-Dilmé, J. Barberà-Rodríguez, S. F. E. Oliviero, **A. A. Mele**. *Strong unitary designs in optimal depth and space*. [arXiv:2608.13491](#) (2026).
7. ★ **A. A. Mele**, Y. Herasymenko. *Efficient learning of quantum states prepared with few fermionic non-Gaussian gates*. *PRX Quantum* 6, 010319 (2025).
8. ★ L. Bittel, J. Eisert, L. Leone, **A. A. Mele**, S. F. E. Oliviero. *A complete theory of the Clifford commutant*. *Quantum* 10, 2171 (2026).
9. ★ **A. A. Mele**. *Introduction to Haar measure tools in quantum information: A beginner's tutorial*. *Quantum* 8, 1340 (2024).
10. ★ L. Bittel, **A. A. Mele**, J. Eisert, L. Leone. *Optimal trace-distance bounds for free-fermionic states: Testing and improved tomography*. *PRX Quantum* 6, 030341 (2025).
11. L. Bittel, J. Eisert, W. Gong, **A. A. Mele**, L. Schatzki. *Adaptivity is all you need: Optimal stabilizer learning using just single-copy measurements*. [arXiv:2610.02031](#) (2026).
12. M. Yaghubi Rad, R. Puig, S. Hassanandani, L. Coffman, J. Eisert, C. Bravo-Prieto, **A. A. Mele**. *Testing quantum Gaussianity with constant sample complexity*. [arXiv:2609.40270](#) (2026).
13. C. Bravo-Prieto, W. Gong, **A. A. Mele**. *Quantum memory advantage for quantum process tomography*. [arXiv:2607.13476](#) (2026).
14. A. Angrisani, **A. A. Mele**, M. S. Rudolph, M. Cerezo, Z. Holmes. *Simulating quantum circuits with arbitrary local noise using Pauli propagation*. *PRX Quantum* 7, 020313 (2026).
15. D. Miller, K. Levi, L. Postler, A. Steiner, L. Bittel, G. A. L. White, Y. Tang, **A. A. Mele**, et al. *Experimental measurement and a physical interpretation of quantum shadow enumerators*. *Physical Review Research* 8, 023318 (2026) Editor suggestion.

16. A. E. Deneris, P. Braccia, P. Bermejo, N. L. Diaz, **A. A. Mele**, M. Cerezo. *Analyzing the free states of one quantum resource theory as resource states of another*. [Advanced Quantum Technologies](#) 9, e00702 (2026).
17. M. Fanizza, V. Iyer, J. Lee, **A. A. Mele**, F. A. Mele. *Efficient learning of bosonic Gaussian unitaries*. [arXiv:2510.05531](#) (2025).
18. L. Bittel, F. A. Mele, J. Eisert, **A. A. Mele**. *Energy-independent tomography of Gaussian states*. [arXiv:2508.14979](#) (2025).
19. N. L. Diaz, **A. A. Mele**, P. Bermejo, P. Braccia, A. E. Deneris, M. Larocca, M. Cerezo. *A unified approach to quantum resource theories and a new class of free operations*. [arXiv:2507.10851](#) (2025).
20. P. Bermejo, P. Braccia, **A. A. Mele**, N. L. Diaz, A. E. Deneris, M. Larocca, M. Cerezo. *Characterizing quantum resourcefulness via group-Fourier decompositions*. [arXiv:2506.19696](#) (2025).
21. L. Bittel, **A. A. Mele**, J. Eisert, L. Leone. *PAC-learning of free-fermionic states is NP-hard*. [Quantum](#) 9, 1665 (2025).
22. L. Bittel, F. A. Mele, **A. A. Mele**, S. Tirone, L. Lami. *Optimal estimates of trace distance between bosonic Gaussian states and applications to learning*. [Quantum](#) 9, 1769 (2025).
23. M. West, **A. A. Mele**, M. Larocca, M. Cerezo. *Real classical shadows*. [Journal of Physics A: Mathematical and Theoretical](#) 58, 245304 (2025).
24. J. Liu, F. Wilde, **A. A. Mele**, X. Jin, L. Jiang, J. Eisert. *Stochastic noise can be helpful for variational quantum algorithms*. [Physical Review A](#) 111, 052441 (2025).
25. M. West, **A. A. Mele**, M. Larocca, M. Cerezo. *Random ensembles of symplectic and unitary states are indistinguishable*. [arXiv:2409.16500](#) (2024).
26. J. Denzler, **A. A. Mele**, E. Derbyshire, T. Guaita, J. Eisert. *Learning fermionic correlations by evolving with random translationally invariant Hamiltonians*. [Physical Review Letters](#) 133, 240604 (2024).
27. J. J. Meyer, M. Mularski, E. Gil-Fuster, **A. A. Mele**, F. Arzani, A. Wilms, J. Eisert. *Exploiting symmetry in variational quantum machine learning*. [PRX Quantum](#) 4, 010328 (2023).
28. **A. A. Mele**, G. B. Mbeng, G. E. Santoro, M. Collura, P. Torta. *Avoiding barren plateaus via transferability of smooth solutions in Hamiltonian variational ansatz*. [Physical Review A](#) 106, L060401 (2022).

Teaching & Invited Lectures

- 05/2026 **Invited Lecturer**, EPFL Bernoulli Center, Lausanne, Switzerland. *Quantum Learning Theory for Bosonic and Fermionic Systems* (1.5-hour blackboard tutorial), delivered as part of the program *Intersection of Quantum Learning Theory, Quantum Sensing, and Quantum Verification*. [\[program\]](#) [\[notes\]](#) [\[video\]](#)
- 01/2026 **Invited Lecturer**, 2026 QISCA Winter School. Lecture on *Learning t -doped Fermionic and Bosonic Gaussian States*, as part of [Quantum Learning Theory for Bosonic and Fermionic Systems](#). [\[course\]](#) [\[notes\]](#) [\[video\]](#) [\[fermionic paper\]](#) [\[bosonic paper\]](#)
- 10/2025 **Invited Lecturer**, AIMS Workshop and School on *The Theory of Quantum Learning Algorithms*, Cape Town, South Africa. Four-hour tutorial on *Haar Integration Tools in Quantum Information*. [\[program\]](#) [\[Haar tutorial\]](#)
- 08/2025 **Invited Lecturer**, QMATH Masterclass, University of Copenhagen (6 lectures). *Representation Theory in Quantum Information Science*. 3 lectures on representation theory (original lecture notes prepared for the masterclass: [PDF](#)); 3 lectures on Haar measure and Clifford commutant methods, based on [Quantum](#) (2024) and [arXiv:2504.12263](#). ([Masterclass link](#))
- 2025 **MSc Thesis Mentor**, Diego Polimeni. *Rigorous End-to-End Fault-Tolerant Quantum Computation with Minimal Assumptions*.
- 2022–2023 **Teaching Assistant**, *Quantum Information Theory*, Freie Universität Berlin. Led exercise sessions and tutorials; [\[course materials\]](#).

Academic Service

- Program Committees PC member for [QIP 2025](#), [QTML 2025](#), [QCTiP 2026](#), [QTML 2026](#) and (forthcoming) [QIP 2027](#).

Refereeing Referee for QIP, TQC, QTML, and QCTiP over multiple years, as well as for physics journals.

Research visits

- 03/2026 **California Institute of Technology (Caltech)**, Pasadena, USA. Hosted at IQIM by Hsin-Yuan (Robert) Huang and John Preskill.
- 02/2026 **Massachusetts Institute of Technology (MIT)**, Cambridge, USA. Hosted at CTP by Soonwon Choi.
- 02/2026 **Harvard University**, Cambridge, USA. Hosted at HQI by Susanne Yelin.
- 05/2025 **Perimeter Institute for Theoretical Physics**, Waterloo, Canada. Hosted by Farzin Salek.

Talks

Selected conference contributions

- 01/2026 *A complete theory of the Clifford commutant and applications*. QIP 2026.
- 01/2026 *Efficient Learning Algorithms for Structured Bosonic and Fermionic Unitary Operators*. QIP 2026.
- 09/2025 *Pauli Propagation: A computational framework for simulating quantum systems*. QTML 2025.
- 09/2025 *A complete theory for the Clifford commutant*. TQC 2025.
- 02/2025 *Learning and testing quantum states of fermionic systems*. QIP 2025.
- 02/2025 *Tomography of bosonic systems and optimal estimates of the trace distance between Gaussian states*. QIP 2025.
- 04/2025 *Tomography of bosonic systems and optimal estimates of the trace distance between Gaussian states*. QCTiP 2025.
- 09/2024 *Learning quantum states of continuous variable systems*. QTML 2024.
- 09/2024 *Learning quantum states of continuous variable systems*. TQC 2024.
- 09/2024 *Noise-induced shallow circuits and absence of barren plateaus*. TQC 2024.
- 08/2024 *Learning quantum states of continuous variable systems*. AQIS 2024.
- 08/2024 *Learning and testing possibly magical fermions*. AQIS 2024.
- 08/2024 *Noise-induced shallow circuits and absence of barren plateaus*. AQIS 2024.
- 11/2022 *Avoiding barren plateaus via transferability of smooth solutions in Hamiltonian Variational Ansatz*. QTML 2022 (extended talk).
- 09/2022 *Avoiding barren plateaus via transferability of smooth solutions in Hamiltonian Variational Ansatz*. DPG 2022.
- 09/2021 *Exploiting symmetry in variational quantum machine learning*. DPG 2021.

Selected invited talks

- 10/2026 Invited Speaker (forthcoming), [Quantum Innovators Theory Workshop 2026](#), Institute for Quantum Computing, University of Waterloo. Workshop bringing together early-career researchers making an impact on quantum information research.
- 05/2026 *Quantum Learning Theory for Bosonic and Fermionic Systems*, EPFL Bernoulli Center (program on Intersection of Quantum Learning Theory, Quantum Sensing, and Quantum Verification). ([link](#))
- 03/2026 *Optimal learning of quantum channels in diamond distance*, Caltech IQIM Seminar.
- 02/2026 *Optimal learning of quantum channels in diamond distance*, MIT Quantum Seminar.
- 02/2026 *Optimal learning of quantum channels in diamond distance*, Harvard Quantum Seminar.
- 09/2025 *A complete theory for the Clifford commutant*, Los Alamos National Laboratory.
- 08/2025 *Invited lecturer at the QMATH Masterclass on Representation Theory in Quantum Information Science* ([masterclass link](#)), University of Copenhagen.
- 08/2025 **TEDx Policoro**. *Quantum Computer: una possibile rivoluzione*, with Francesco A. Mele. ([video](#)) ([event](#))
- 07/2025 *Effect of noise in typical quantum circuits*, *Understanding Quantum Machine Learning* workshop, Galileo Galilei Institute, Florence, Italy.
- 05/2025 *Learning and testing quantum states of fermionic systems*, Perimeter Institute for Theoretical Physics, Waterloo, Canada.

- 05/2025 *Effect of noise in typical quantum circuits*. The Higgs Centre for Theoretical Physics, Edinburgh, UK.
- 04/2025 *Effect of noise in typical quantum circuits*. Scuola Normale Superiore (SNS), Pisa, Italy.
- 09/2024 *Noise-induced shallow circuits and absence of barren plateaus*. Quantinuum.
- 04/2024 *Noise-induced shallow circuits and absence of barren plateaus*. Centre for Quantum Technologies, NUS.
- 04/2024 *Noise-induced shallow circuits and absence of barren plateaus*. BraQIIT Lab, IIT-Delhi.

Schools and Workshops

- 08/2025 *QMATH Masterclass on Representation Theory in Quantum Information Science*, University of Copenhagen, invited lecturer.
- 07/2025 *Workshop on the Clifford commutant*, Cologne. Organized to discuss our [work](#).
- 07/2025 *Understanding Quantum Machine Learning*, Galileo Galilei Institute, Florence, invited speaker.
- 05/2025 *Entanglement, Information and Complexity in Quantum Systems*, Edinburgh, invited speaker.
- 06–08/2024 *Quantum Computing Summer School*, Los Alamos National Laboratory (acceptance ratio < 3%).
- 2022–2024 QMATH Masterclasses: Entropy Inequalities (2022), Quantum Learning Theory (2023), and Quantum Simulation (2024), University of Copenhagen.
- 07/2023 *PCMI Graduate Summer School in Quantum Computation*, IAS/Park City.

Awards & Fellowships

- 2027–Present **Simons Institute–Jane Street Circles Program**. Collaborative quantum algorithms project selected for funding in the Circles program co-hosted by the Simons Institute for the Theory of Computing and Jane Street, supporting four weeks of in-person research collaboration over two years.
- 2025–Present **Google PhD Fellowship in Quantum Computing** (US\$70,000/year in research support). Awarded by Google to “recognize outstanding graduate students doing exceptional and innovative research in areas relevant to computer science and related fields.”
- 2019 **University of Trento–SISSA MSc Scholarship**. One of seven recipients, selected through a competitive examination in physics and mathematics. Included financial support and access to advanced PhD-level courses at International School for Advanced Studies (SISSA).

Earlier Achievements

- 2015 **Italian Informatics Olympiad**. National finalist.
- 2015 **IoStudio Award** (MIUR & Samsung). Winner for developing an Android application for my high school.
- 2014–2016 **Elected Student Body Representative**, Liceo Scientifico “Enrico Fermi” (Policoro, Italy) — two consecutive years; public speaking and organizing high school workshops and student activities.
- MSc thesis *Quantum Approximate Optimization Algorithm for Quantum Many-Body States Preparation*. Advisors: Mario Collura; Philipp H.J. Hauke.
- BSc thesis *How to break down quantum noise in a gravitational waves detector*. Advisor: Giancarlo Cella.

Theory, AI & Scientific Computing

- Theory Quantum information, quantum algorithms and complexity; broad mathematical background, with particular expertise in representation theory, probability theory, matrix analysis, and random matrix theory, and experience rapidly learning and applying new mathematical tools.
- AI & ML Learning theory and statistical inference, with particular emphasis on quantum learning; large language models for mathematical reasoning, scientific programming, and AI-assisted research.
- Computing Python, Mathematica, C/C++; scientific computing and numerical linear algebra; algorithm implementation and numerical benchmarking; quantum-circuit and many-body simulation; Cirq, Qiskit, PennyLane.